

Article

Model for Predicting Maize Crop Yield on Small Farms Using Clusterwise Linear Regression and GRASP

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Abstract: Planting a crop involves several key steps: resource assessment, crop selection, crop rotation, planting schedules, soil preparation, planting, care, and harvesting of crops. In this context, estimating the productivity of a crop based on available information, such as expected climatic conditions and agricultural practices, helps farmers reduce the uncertainty of their investment. In Colombia, maize is the fourth most important crop in the country. Significant efforts are required to improve productivity in traditional and technified production systems. In this sense, this research proposes and evaluates an approach called Clusterwise Linear Regression (CLR) to predict the crop maize yield in small farms, considering data on climate, soil, fertilization, and management practices, among others. To develop the CLR model, we conducted the following steps: data collection and preparation, clustering using k-means, cluster optimization with Greedy Random Adaptive Search Procedure (GRASP), and performance evaluation. The cluster optimization process allows the identification of clusters with similar characteristics and generates multiple linear regression models with mixed variables that explain the yield of the farms on each cluster. The Simulated Multiple Start Annealing (MSSA) metaheuristics were also evaluated, but the results of GRASP were the best. The results indicate that the proposed CLR approach is more effective than the linear and nonlinear algorithms mentioned in the literature, such as multiple lasso linear regression, random forests, XGBoost, and support vector machines. These algorithms achieved an accuracy of 70%. However, with the new CLR model, a significantly improved accuracy of 87% was achieved with test data. The clusters' studies revealed key factors affecting crop yield, such as fertilization, drainage, and soil type. This transparency is a benefit over black-box models, which can be harder to interpret. This advancement can allow farmers to make better decisions about the management of their crops.

Keywords: clusterwise; linear regression; metaheuristics; clustering; site-specific agriculture

MSC: 90C59; 68T20; 62-07



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1. Introduction

Maize is considered the second crop in the world in terms of production, second only to wheat, and is of great economic importance worldwide due to its wide variety of uses as human food, feed for animal species, or as raw material for the transformation of starch into different products for industries such as papers, textiles, pharmaceuticals, and adhesives, among others [1,2]. Maize is Colombia's third most demanded product after coffee and rice. This crop represents 12% of the country's planted area; it is the fourth most important in terms of planted area, after coffee (16%), rice (12%), and oil palm (12%), where numerous rural families are dedicated to this activity, and it is also considered a source of income and work in several regions of Colombia [3]. In Colombia, there are currently two main production systems for maize. The technified system is often

characterized by monocultures with more than 5 Ha (hectares) that have all the resources and technologies based on mechanization for soil preparation and use of improved seeds, among others. This system represents 55.4% of the area destined for maize production, with a 1.512 Mt (megatons) production and an average yield of 5.90 Tn/Ha for 2022. The traditional system uses native seeds and does not employ mechanized technology. In 2022, this system produced 0.408 Mt with an average yield of 1.98 Tn/Ha. On the other hand, the technified system uses improved seeds and mechanized technologies throughout the cultivation process, resulting in a 2022 production of 1.512 Mt and an average yield of 5.90 Tn/Ha [4]. According to data from the National Federation of Producers of Cereals, Legumes, and Soybeans (FENALCE), the total amount of maize imported in 2022 was 6.49 Mt, while the total national production was 1.921 Mt.

Accurate prediction of crop yields is crucial for optimizing agricultural production and assisting farmers in improving their life conditions. Although Colombia has made considerable progress in using site-specific agriculture (SSA) to improve production levels, there is still room for improvement by implementing ML algorithms. ML algorithms will help maize producers better understand crop management, reduce incorrect use of resources, lower production costs, and improve competitiveness in the international market [5]. It is here where the SSA plays a crucial role because it allows agronomic techniques to be conducted according to the spatial and temporal conditions of the site where the crop is grown and to obtain the highest yield.

From the above, studies have emerged focusing efforts on understanding the most influential elements or decisions in crops, indicating that every time a farmer makes a planting, there is a unique event; it is an experiment with virtually no history, which can be monitored throughout its growth and production. By recording many of these events and implementing a traditional statistical modeling approach or artificial intelligence-based modeling, farmers can be provided with information on choosing techniques for appropriately managing their crops [6]. Significant contributions have been made in the area focused on predicting maize crop yield; some of them focused on the use of structured data and making use of a wide variety of Machine Learning (ML) methods, such as in 2022 [7], where the spatial model Geographically Weighted Random Forest Regression (GWRFR) was presented to predict maize production at the county level in the U.S. maize belt. This study uses different datasets, such as climate data, soil data, and vegetation indices, among others. The GWRFR model was compared against five popular machine learning algorithms: Multiple Linear Regression (MLR), Partial Least Squares Regression (PLSR), Support Vector regression (SVR), Decision Tree Regression (DTR), and Random Forest Regression (RFR). The results show that the GWRFR model achieves an adjustment of ($R^2 = 0.90$ and $RMSE = 0.764$ MT/ha), which significantly exceeds the results obtained with the other models.

Also, in 2022 [8], the ABSOLUT algorithm was developed using multiple linear regressions for each spatial subunit; this model uses the full spectrum of meteorological averages in different spatial time windows. The algorithm's results were optimal compared to linear regressors and decision trees, among others. However, the authors conclude that it did not consider other factors affecting yield, such as extreme weather, management practices, and fertilization; therefore, it is not possible to generalize the performance of the model. In 2021 [9], a model called GEM (Genetics, Environment, and Crop Management) was presented, which was used to predict maize yield at each country's level in the United States. The performance metric used to evaluate the quality of the solution was the RMSE, where good results were achieved compared to other ML algorithms, but it is concluded that the model cannot be generalizable for all counties; in addition, some variables, such as irrigation, fertilization, tillage, and weed control, were not considered.

Other research in [10] (2019), [11] (2020), and [12] (2022) proposed using different ML algorithms, such as highly randomized trees, gradient boosting machines, XGBoost, and algorithm artificial neural networks (ANNs), using several input variables such as topographic humidity index, topographic position index, spectral bands, vegetation indexes,

and precipitation indexes, among others, to predict maize yield. Also, in [13] (2022), [14] (2022), and [15] (2023), more complex algorithms are proposed using unstructured data (images) captured by satellites, where Deep Learning models are used to predict the vegetation index that is directly related to crop yield, or in [16] (2022), a hybrid approach is developed using the World Food Studies model (WOFOST) to build a dataset and a Deep Learning model to forecast the potential yield of maize by analyzing each growth phase.

Moreover, there are several studies that have applied clustering techniques for the identification of best agricultural management practices based on groupings by climate and soil properties regardless of the geographic location where different crops, such as peanuts and maize, have been planted [17,18] (2021 and 2024). With these cluster analyses, it has been possible to characterize the environment where these crops are grown and identify the major contributing factors within each detected grouping. For example, in [19] (2017), a clustering model was proposed using only soil properties, including phosphorus (p), potassium (K₂O), electrical conductivity, pH value, organic carbon, and soil texture, to predict wheat yield. The results show that several similarities are shared in many events that are not detectable to the naked eye; however, by applying clustering, it is possible to identify similarities between lands. Furthermore, in [20] (2018), fuzzy c-means were used with a neural network to predict wheat yield, considering the clustering of environmental variables such as temperatures, precipitation, and biomass.

On the other hand, different optimization algorithms have also been used to maximize crop yields. Some of them are bioinspired, like in [21–23], or others are based on physical principles. These algorithms have provided solutions to different complex problems, such as sowing planning, crop rotations, and optimization of water resources, among others [24,25].

Despite the progress made in the area, there are still challenges regarding model accuracy and interpretability. In general, crop yield prediction can be influenced by a myriad of factors, such as excess nitrogen, crop drainage, climatic variation, seed quality (genotype), lack of crop rotation, and lack of technical support, among others [26–29]. Therefore, it is necessary to search for new approaches that not only achieve high precision but also allow understanding the effect and influence of each of the predictors on the target variable (yield).

Considering the data generally recorded in agriculture are heterogeneous and vary according to the specific conditions of each site. In this research, a contribution from computer science to SSA has been achieved by integrating the Clusterwise approach, which allows grouping data into clusters and defining regression models within each cluster simultaneously [30,31]. Clusterwise is a combination of two methods: clustering and regression. This approach has been implemented in several areas of applications, such as prediction and optimization of hospital demand [32], rainfall prediction [33] (2017), accident frequency prediction [34] (2021), and pavement management, where excellent results have been obtained for road management [35,36] (2017 and 2020). For the case of SSA for short-cycle crops, a great opportunity to implement the CLR approach was evidenced due to many explanatory variables being used, and their contributions to the response variable vary from one specific site to another.

This research aimed to introduce a new algorithm that defines a CLR model for the prediction of maize yield. This algorithm uses various input data variables related to climate, soil, fertilization, and management practices. It employs the k-means algorithm and an optimization process using an adaptation of the Greedy Randomized Adaptive Search Procedures (GRASP) [37] or Multi-Start Simulated Annealing (MSSA) [38] metaheuristics, selecting GRASP as the best option for the optimization process. The algorithm identifies clusters of data with similar characteristics that expose relationships not detectable by traditional algorithms, and those clusters have associated linear regression models that allow crop yields to be predicted accurately. Linear regressions were selected, prioritizing obtaining an interpretable model, which, unlike other complex algorithms such as ensembles, support vector machines, or neural networks, is challenging to interpret.

To date, the authors of this research have yet to find a state-of-the-art report on the use of CLR for crop yield prediction, so it is vital to make this technique known, as it is efficient in performance and interpretability. The main contributions of this research are: (1) To propose a new algorithm to define a CLR model that predicts the yield of maize crops with a performance superior to other machine learning techniques; (2) To propose an interpretable CLR model that allows experts and farmers to understand how the model predicts the yield of their crops, thus improving their agricultural practices; (3) To show how to adapt the MSSA and GRASP metaheuristics to optimize an initial CLR model built with the k-means algorithm and linear regression models for maize crop yield prediction; (4) To evaluate and compare the performance of the proposed CLR model against other state-of-the-art algorithms; and (5) Perform a descriptive analysis of the CLR model obtained, identifying management practices applicable to Colombian farmers.

This article is organized as follows: Section 2 provides an overview of the materials and methods used in the research. It includes a description of the datasets, their attributes, and the transformations applied to the dependent and independent variables. Additionally, it describes the CLR approach and the adaptations made to MSSA and GRASP metaheuristics to find a CLR solution. Section 3 shows the results obtained, presenting a comparison between the results of the two (2) metaheuristics, i.e., MSSA and GRASP. Also, it analyzes the performance of the CLR model against other machine learning algorithms and presents a detailed analysis of the best selected model. Finally, Section 4 presents the work's conclusions and outlines future research.

2. Materials and Methods

2.1. Study Area

The study data were collected from maize crops from the Department of Cordoba in Northern Colombia; this region is characterized by its climate and varied relief suitable for producing various products such as maize, wheat, and bananas. It is considered one of the leading maize producers in the country, contributing 12.26% of national production in 2022 (2nd in production of technified white maize, 5th in production of technified yellow maize, and 9th in production of traditional yellow or white maize) [39]. The relief of its towns is highly heterogeneous, which allows for variable production levels each year. Figure 1 shows the topographic location of the department and the average yield for traditional maize between 2016 and 2018.

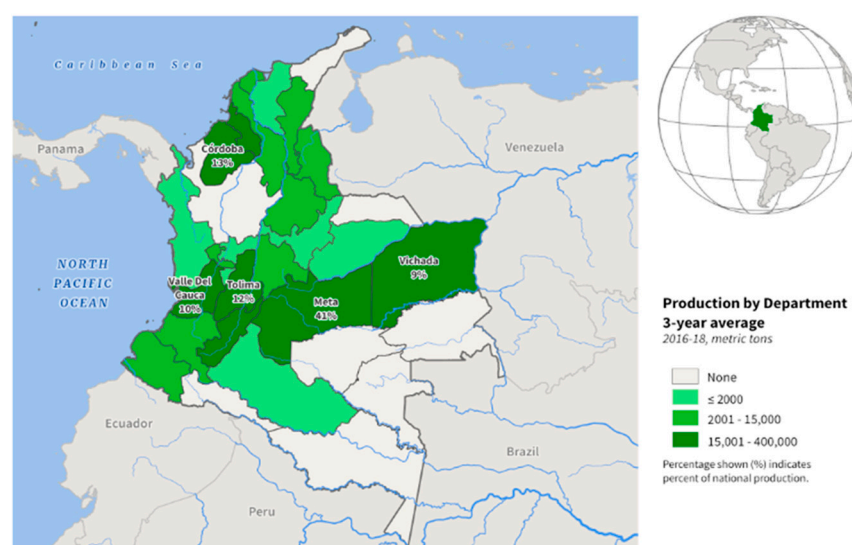


Figure 1. Average yield maize production for the Cordoba department between 2016 and 2018, taken from the U.S. Department of Agriculture [40].

2.2. Dataset

The input dataset was collected from monitoring maize crops within the department of Cordoba, Colombia, between 2013 and 2017. The records contain information corresponding to agricultural management practices, soil properties, and climatic information.

The data collection process was expertly conducted by FENALCE agronomists, who visited the crop areas to gather information from the farmers. This information was then centralized in the SIRIA virtual platform [41], which is managed by FENALCE. The collected information includes 813 records on field characterization, captured by farmers and technicians following the RASTA (Rapid Soil and Terrain Assessment) guide [42]. These records provide details on the area planted, inclination level, location, and land characterization used in planting, as per the guide's procedure. Table 1 presents some of the variables and their definitions captured by FENALCE agronomists.

Table 1. Variables collected for characterizing maize crops in the Cordoba department.

Variable	Meaning	Type	Unit
Id Lot	Lot identifier where the planting took place	Continuous	-
Planting Type	Planting can be performed manually or mechanized.	Categorical	-
Treated Seeds	Variable that allows determining if the seeds have been treated before planting	Binary	-
Genetic Material	Type of genetic material associated with the seed: DNA or RNA.	Categorical	-
Previous Crop	Variable describing the type of crop previously planted in the field.	Categorical	-
Drainage	Determine if the soil has good drainage (YES, NO)	Binary	-
Harvesting method	It determines how the product was harvested; it can be manual or mechanized.	Categorical	-
On-farm storage	This variable indicates whether the crop was stored on the producer's farm	Binary	-
Days to emerge	Days it took for the seed to emerge since it was planted in the soil	Continuous	-
Days in Emerging to flowering	The number of days that have passed between the emergence and flowering stages of the plants.	Continuous	-
Days in flowering to harvest	The number of days that have passed between the flowering and harvest stages of the plants.	Continuous	-
Population 20 Days	Number of plants, 20 days after Planting	Continuous	-
Lat Lot	Latitude Lot	Continuous	°
Long Lot	Longitude Lot	Continuous	°
Furrow distance	The Furrow distance variable refers to the spacing between furrows in agricultural fields.	Continuous	cm
Humidity	Refers to the water content present in the soil	Continuous	%
Field Type	Refers to the characteristics of the field such as: undulating, mountainous, or flat.	Categorical	-
Number of soil layers	Variable that identifies the number of layers in the soil where the planting was conducted.	Categorical	-
Thickness	Refers to the thickness associated with each of the soil layers (cm)	Continuous	cm

Table 1. *Cont.*

Variable	Meaning	Type	Unit
Soil Texture	It refers to the distribution of particles of varied sizes that comprise the soil, such as sandy soil, silty soil, and clay soil	Categorical	-
PH soil	Soil acidity level	Continuous	-
Yield	Crop productivity	Continuous	Kg-Ha

The original dataset is available at [43]. The climatic data were generated using the geolocation coordinates associated with the crops, planting dates, and data from local weather stations obtained from the National Institute of Meteorology IDEAM. This process was defined by CIAT (International Center for Tropical Agriculture). Through this process, we were able to estimate the average temperature, radiation levels, and precipitation percentages during the stages of the maize crops (vegetation, formation, and maturity). Table 2 describes the variables associated with the climate used to construct the mineable view.

Table 2. Climate variables used in the construction of the CLR approach.

Variable Name	Meaning	Type	Unit
Temp_Max_Avg_Veg	Average maximum temperature in vegetative stage	Continuous	°C
Temp_Min_Avg_Veg	Average minimum temperature in vegetative stage	Continuous	°C
Temp_Avg_Veg	Average temperature in vegetative stage	Continuous	°C
Diurnal_Range_Avg_Veg	Average temperature Diurnal Range in vegetative stage	Continuous	°C
Sol_Ener_Accu_Veg	Accumulated solar energy in vegetative stage	Continuous	Cal-cm ²
Temp_Max_34_Freq_Veg	frequency of days with maximum temperature above 34 °C in vegetative stage	Continuous	-
Rain_Accu_Veg	Accumulated precipitation in vegetative stage	Continuous	mm
Rain_10_Freq_Veg	Frequency of days with more than 10 mm precipitation in vegetative stage	Continuous	-
Rhum_Avg_Veg	Average relative humidity in vegetative stage	Continuous	%
Temp_Max_Avg_For	Average maximum temperature in formation stage	Continuous	°C
Temp_Min_Avg_For	Average minimum temperature in formation stage	Continuous	°C
Temp_Avg_For	Average temperature in formation stage	Continuous	°C
Diurnal_Range_Avg_For	Average temperature Diurnal Range in formation stage	Continuous	°C
Sol_Ener_Accu_For	Accumulated solar energy in formation stage	Continuous	Cal-cm ²
Temp_Max_34_Freq_For	frequency of days with maximum temperature above 34 °C in formation stage	Continuous	-
Rain_Accu_For	Accumulated precipitation in formation stage Continuous mm	Continuous	mm
Rain_10_Freq_For	Frequency of days with more than 10 mm precipitation in formation stage	Continuous	-
Rhum_Avg_For	Average relative humidity in formation stage	Continuous	%
Temp_Max_Avg_Mad	Average maximum temperature in maturity stage	Continuous	°C
Temp_Min_Avg_Mad	Average minimum temperature in maturity stage	Continuous	°C
Temp_Avg_Mad	Average temperature in maturity stage	Continuous	°C

Table 2. Cont.

Variable Name	Meaning	Type	Unit
Diurnal_Range_Avg_Mad	Average temperature Diurnal Range in maturity stage	Continuous	°C
Sol_Ener_Accu_Mad	Accumulated solar energy in maturity stage	Continuous	Cal-cm ²
Temp_Max_34_Freq_Mad	frequency of days with maximum temperature above 34 °C in maturity stage	Continuous	-
Rain_Accu_Mad	Accumulated precipitation in maturity stage	Continuous	mm
Rain_10_Freq_Mad	Frequency of days with more than 10 mm precipitation in maturity stage	Continuous	-
Rhum_Avg_Mad	Average relative humidity in maturity stage	Continuous	%

With the combination of these information sources, the minable view was constructed for use in the modeling and definition of the CLR approach [44].

2.3. Data Preprocessing

All the variables included in minable view were treated with data preprocessing standard techniques for data mining projects, such as outlier detection based on the Interquartile range, analysis of counts by class, categorical variable encoding, and data imputation. However, it should be mentioned that in the present work, there were situations where some data had to be specially processed, and these cases are described as follows:

Adjustment to columns with dependent values: In certain columns, a unique value was assigned due to some cases where there was a correlation between the variables. This was noticed in the data related to the RASTA guide, where a high correlation was found between the variables linked to the presence of rocks, stones, and roots in the soil and the depth at which they were located. When data were absent in the first column in the dependent columns, no value was recorded, which caused the column to appear to have missing data, which was not correct, so it was necessary to identify in which cases there should be a value instead of a null value, thus allowing to know that it was a particular situation and not the absence of data. It was decided to replace the null value with a constant, in this case, -1 , because it differed from those already stored in the attribute (numeric values equal to or greater than zero) and was easily identified as an imputed value.

Division by percentage values: A sample of the RASTA dataset with variables representing soil color, texture, breaking strength, and organic material was reported in Table 3. These values varied in the number of observations stored for each record due to the different number of layers that could be observed when opening the box in the soil, as suggested by the guide. By not having the same number of observations (layer) and changing the size of each layer (measured in cm), a process of division of variables into several columns based on percentage values had to be developed.

Table 3. The initial representation of variables associated with texture.

ID_LOT	NO_LAYERS_RASTA	THICKNESS	TEXTURE
40	3	22, 9, 50	FAr, FrL, FA
43	3	20, 14, 36	FAr, FAr, ArL
44	3	26, 14, 38	FAr, FAr, AF
45	3	16, 39, 29	FAr, FrL, FrL
46	3	33, 6, 47	FAr, FAr, FrL
47	3	10, 20, 24	FAr, FAr, FrL
51	3	18, 33, 30	FAr, ArA, Ar

This process consists of creating N new columns for each of the variables with this characteristic, and each of these N columns represents one of the possible values that could

exist within the variable. The depth of the box represents the maximum possible thickness value that a layer could take and corresponds to 100%; then, for each of the reported values together with its observed thickness, the incidence of this observation inside the box was calculated in terms of percentage and stored in its respective column as can be seen in Table 4, where the texture FAr, FrL, and FA of field 40 are transformed to values in the columns Por_FAr, Por_FrL, and Por_FA equal to 27.16, 11.11, and 61.73, respectively, based on the thickness of each layer (22, 9, and 50).

Table 4. The final representation of variables associated with texture.

ID_LOT	Porc_A	Porc_Ar	Porc_ArA	Porc_ArL	Porc_FrL	Porc_L	Porc_F	Porc_FAr	Porc_FA	Porc_Af
40	0	0	0	0	11.11	0	0	27.16	61.73	0
43	0	0	0	51.43	0	0	0	48.57	0	0
44	0	0	0	0	0	0	0	51.28	0	48.72
45	0	0	0	0	80.95	0	0	19.05	0	0
46	0	0	0	0	54.65	0	0	45.35	0	0
47	0	0	0	0	44.44	0	0	55.56	0	0
51	0	37.04	40.74	0	0	0	0	22.22	0	0

Date transformation: The dates recorded in the files alone did not provide much information for the data mining algorithms, so they were used to generate knowledge of the number of days elapsed between crop stages: planting—germination, germination—flowering, and flowering—harvest, and thus provide more information to the algorithms.

On the other hand, the dates were also used in conjunction with the data obtained on fertilization and controls; for this purpose, the types of fertilizations and controls conducted were identified, and the repetitions of these types of events were counted within the time elapsed between the crop stages. The nitrogen, potassium, and phosphorus content related to fertilization were also added.

Transformation of the yield variable: The column of the minable view contained the original weight after harvest, but the final weight of the grain is related to humidity. According to the FENALCE technicians, the percentage of humidity lost by the grain after harvest is 14%; therefore, to obtain the new values, the recorded yield variables were used, the humidity contained by the grain at the time of harvesting, and with these variables, Equation (1) was used to define the new yield in kg/ha.

$$RDT_{Adjusted} = \frac{100 - Humidity}{100 - 14} \times Yield \quad (1)$$

2.4. Dimensionality Reduction

With the selected and processed set of variables of climate, soil, and management practices integrated into a single file, there were 799 records and 115 variables. For a file with this number of columns, reducing the number of features (variables) is recommended since, in this way, the algorithms would reduce their processing time. Irrelevant, noisy, and highly correlated characteristics that affect the algorithms' performance should be eliminated (removed) [45]. There are multiple techniques to reduce the dimensionality of variables by generating new ones (attribute engineering), such as principal component analysis (PCA), singular value decomposition (SVD), and linear discriminant analysis (LDA), among others. However, feature selection techniques were analyzed for this work to preserve the original features. The methods evaluated were high correlation, wrapper, and low variance. Each of them is described as follows:

2.4.1. High Correlation

This method analyzes all variables (115) and detects the highly correlated ones (values between -1.0 and -0.8 or between 0.8 and 1.0). However, it was problem-

atic in several cases because there are correlations such as Temp_Max_Avg_Mad and Temp_Max_34_Freq_Mad where it was not possible to define which feature to eliminate and which to keep without the knowledge or advice of an expert to explain the influence of these on maize crops. In addition, there were cases where several pairs of characteristics had a strong correlation, as is the case shown in Table 5, where the group of variables Temp_Max_Avg_Veg, Temp_Max_34_Freq_Veg, and Temp_Avg_Veg shows a correlation of 92.8% for the first two, 82.5% for the last two, and finally 89.9% for the first one with the last one. In these cases, it was impossible to decide which variables to keep and which to discard without the advice of an expert. Therefore, the option of reducing the number of variables based on the correlation between them was discarded.

Table 5. Correlation between climate variables.

First Attribute	Second Attribute	Correlation
TotN_Antes_Siem	FerQui_Antes_Siem	0.975
Temp_Max_Avg_Mad	Temp_Max_34_Freq_Mad	0.956
Temp_Max_Avg_For	Temp_Max_34_Freq_For	0.952
Rain_Accu_Veg	Rain_10_Freq_Veg	0.942
Temp_Max_Avg_Veg	Temp_Max_34_Fre_Veg	0.928
Temp_Max_Avg_Veg	Temp_Avg_Veg	0.899
Rain_Accu_For	Rain_10_Freq_For	0.885
Rasta_Slope	External_Drainage	0.872
Temp_Max_Avg_veg	Diurnal_Range_Avg_Veg	0.861
Temp_Max_Avg_For	Temp_Avg_For	0.861
Temp_Min_Avg_Mad	Temp_Avg_Mad	0.854
Rain_Accu_Mad	Rain_10_Freq_Mad	0.829
Temp_Max_Avg_Mad	Temp_Avg_Mad	0.829
Temp_Min_Avg_For	Temp_Avg_For	0.827
Temp_Avg_Veg	Temp_Max_34_Freq_Veg	0.825

2.4.2. Wrapper

Wrappers are feature selection algorithms that, starting from a dataset and a classifier or regressor, select the best combination of features that maximizes a quality criterion (precision, recall, F-measure, or other) and return the best combination and the importance of the features. Within Weka, the WrapperSubsetEval with the GreedyStepwise search method was used to select features.

The wrapper was executed 37 times using different random seed values. In each of these executions, the input dataset was the same, and the obtained lists of variables selected as influential were different. Using these 37 lists, the number of occurrences for each variable was identified; also, it was observed that some variables were chosen in the totality of the executions, and some were never selected.

Once the number of occasions on which the variables were selected was identified, they were ordered from highest to lowest according to the frequency of occurrence; thus, those chosen as relevant on most occasions came first. With the sorted data, the number of occurrences was calculated in terms of percentage; then, a new column was generated where a cumulative percentage was stored.

The column with the accumulated percentages was used to obtain an interpretation of the extent to which a group of variables contributed importance or knowledge to the problem; for example, when selecting the first five variables, it was observed that they had an accumulated percentage of 16.24 as shown in Table 6. Based on this, two new mineable views were obtained, one containing the variables up to where a cumulative percentage of 90% was obtained and the second up to where the cumulative percentage reached 95%.

Table 6. Incidents in the selection of variables by Wrapper.

Variable	Appearances	Percentage	Accumulate	Contribution Approximate	Percentage Cumulative
Harvest_Method	37	100	100	3.25	3.25
Type_Material	37	100	200	3.25	6.49
TotP_Siem_Emer	37	100	300	3.25	9.74
Observe_Moho_Rasta	37	100	400	3.25	12.99
Deep_Living_Roots_Rasta	37	100	500	3.25	16.24
Observe_Plants_Small_Rasta	37	100	600	3.25	19.49
Temp_Min_Avg_Mad	37	100	700	3.24	22.73
Genetic_Material	36	97.3	797.3	3.16	25.89
Sol_Emer_Accu_Veg	36	97.3	894.6	3.17	29.06
Dry_Arid_Region_Rasta	35	94.9	989.2	3.07	32.13
Lot_Heigth	34	91.9	1081.1	2.98	35.11
Rain_Accu_For	34	91.9	1173.0	2.99	38.10
Diurnal_Range_Avg_Veg	33	89.2	1262.2	2.90	41.00
ContMalMec_Flor_Cose	31	83.8	1345.9	2.72	43.72
Rain_10_Freq_Flor	31	83.8	1429.7	2.72	46.44
FerOrg_EmerFlor	30	81.1	1510.8	2.63	49.07
Observes_Crusts_White_Rasta	29	78.4	1589.2	2.55	51.62
Cult_Ant	28	75.7	1664.9	2.46	54.08
Sol_Ener_Accu_Mad	25	67.6	1732.4	2.19	56.27
Profund_Mottled_Rasta	24	64.9	1797.3	2.11	58.38
Rhum_Avg_For	23	62.2	1859.5	2.02	60.40

These two minable views will be used as a test to evaluate the impact of feature selection using wrappers and how it affects or improves the performance of the CLR approach proposed.

2.4.3. Low Variance

Because it was known that the CLR approach would divide the dataset of the mineable view into N clusters, it was suspected that if a variable had a low variance in the final dataset, then dividing these into clusters would result in an even lower or even zero variance in the variable in interest within the resulting clusters. Therefore, we proceeded to eliminate characteristics with low variance, i.e., those where all their records had the same value; we also eliminated those characteristics where there was a small set of values and the proportion of the frequency of the most common value compared to the frequency of the second value was very large, i.e., where there was a dominant presence of a particular value. When running the CLR approach experiments with the dataset without applying variable reduction in any kind, it was identified that some of the variables that were candidates to be eliminated due to their low variance were being selected in the regression models for some clusters, which indicated that within these groups the values were influential and also when splitting the dataset the variance could increase in some cases. For this reason, this strategy was also discarded.

Finally, no feature reduction was performed when constructing the CLR model; the original minable view was used.

2.5. Adaptation of CLR Approach

Starting from the defined minable view, the CLR model is developed, beginning with a process called “initial shuffling” and then with a second process called “optimization”,

where the K clusters are finally obtained. Each cluster contains elements (records or tuples) that best fit a regression model. The R^2 adjusted value measures the fit quality, assuming the models obtained are statistically significant. The initial shuffling is the process responsible for distributing the observations in distinct clusters to generate a starting point for the optimization process, which is performed using the GRASP or MSSA metaheuristics. There was a previously established number K of groups (clusters) to do this. Each cluster has a set of N similar observations associated with it under the distance criterion used and is represented as shown in Figure 2. The identifiers of the instances (record or row) that belong to each group or cluster are recorded.

Cluster 1						Cluster 2						...	Cluster 5					
1	10	799	...	11	51	4	720	8	...	40	32		2	12	622	...	7	614

Figure 2. Representation of clustering observations (example for 5 clusters).

The k-means algorithm [46] was used for the initial shuffling process, using the Euclidean distance criterion for mixed variables because the maize crop data have binary, continuous, and nominal features [47].

After the initial shuffling, the MSSA or GRASP metaheuristic was used to optimize the search space over a determined period, taking the distribution obtained in the initial shuffling as a starting point. The search process was supported by multiple linear regression, which generated regression models for each cluster of instances, taking crop yield as the dependent variable. The models generated for each cluster have an associated R^2 adjusted; then, the mean (average) of the R^2 adjusted obtained in the K models is calculated, and the value obtained is taken as a measure of the quality (fitness) of the solution.

Two feasibility criteria were used to define whether a solution is valid when searching for new solutions. The first is that a regression model can be created for each cluster, and the second one is that each model has a statistical significance value (*p* value) below 0.05, ensuring a significant linear relationship between the independent and dependent variables.

Finally, from the process of searching for an optimal solution, the best CLR model is obtained, consisting of a set of K linear regression models and their corresponding K centroids; this solution is shown in Figure 3; these models are associated with each one of the clusters found in the best solution. These multiple linear regression models describe and identify features that have a level of importance and impact on their respective cluster of instances (crops) under the R^2 adjusted criterion associated with each model.

The models were found to help decision-making, considering that one of the main problems in agriculture is identifying management practices that generate positive effects on crops. Using the CLR approach, it is possible to determine these management practices based on site-specific agriculture (SSAP), where practices that affect a crop's yield were identified and arise from data from other crops with similar conditions.

K1						K2						...	K5					
1	10	799	...	11	51	4	720	8	...	40	32		2	12	622	...	7	614
$y = a + b_1X_1 + b_2X_2 + \cdots + b_pX_p$						$y = a + b_1X_1 + b_2X_2 + \cdots + b_mX_m$							$y = a + b_1X_1 + b_2X_2 + \cdots + b_nX_n$					
$R^2_{adjusted}$						$R^2_{adjusted}$							$R^2_{adjusted}$					
$p\ Value$						$p\ Value$							$p\ Value$					
fitness																		

Figure 3. Representation of a solution using the CLR approach.

2.5.1. Initial Shuffling

The initial shuffling process uses the k-means algorithm to distribute the 799 observations into K clusters. This process is presented in Algorithm 1. The main steps are summarized below.

Initial Assignment: K observations were randomly selected from the dataset to assign their values to the initial centroids of the K clusters (Line 1).

Observations Assignment: The 799 observations are assigned to the K clusters; this assignment is conducted under a criterion of distance to the nearest cluster, i.e., the distance of observation to the centroid of each cluster is calculated, and the one with the shortest distance is chosen. The Euclidean distance was selected to measure the difference in longitude between a centroid and an observation. For the implementation, a list of variables to be ignored within the dataset was used so that they would not be considered in calculating distances (Id_Lot and Yield_Adjusted, for example); see lines (4 to 7).

Centroid update: The centroids of the K clusters are recalculated, considering the observations that conform to the cluster. A centroid has a structure such as an observation, composed of 115 variables, and a control structure for each feature that provides additional information, helping to reduce calculations during its update (see lines 8 to 10).

The representation of a centroid is a vector of 115 variables, the same as the mineable view. For the numeric data type columns, an average of the values the instances have is stored; for the nominal and binary ones, the observed frequency of each instance's values is stored. Figure 4 represents an example of a centroid. Figure 4 shows a centroid that belongs to a cluster that has 200 instances; positions 1 and 115 belong to numerical features (continuous) and contain the sum of the values that the 200 instances have in those features, positions 2, 3, and 4 are binary and nominal variables and therefore have the observed frequency of the values that the 200 instances have; this means that for position 2, 120 observations have the machined value and 80 the manual value. With this representation, when an observation is added or removed in a cluster, it is faster to update the control structures of the 115 centroid variables. The mean is taken for the numerical variables to determine the value that the centroid uses as a representative for each of the 115 variables, and the mode is used for the binary or nominal variables. For the example in Figure 4, the value taken by the centroid at position 2 is mechanized because it corresponds to the mode in that feature, and for position 115, the value to take is 5100.6 ($=1,020,120/200$) because it is the mean of the 200 instances that belong to the cluster.

Finally, at the end of the execution of the k-means algorithm, a set of K clusters of observations is obtained, as shown in Figure 3; this result is validated using the feasibility criteria to ensure that it is a viable (feasible) solution to continue with the optimization process. If the initial distribution of observations in the K clusters is unsuccessful, it is repeated until an initial list of validated clusters is found.

Algorithm 1. K-means Algorithm Initial shuffling process.

Input:	Number of clusters:	k
	Maximum Iterations:	max_iter
	Dataset:	data
	Distance type:	dis_type
	List of variables to ignore:	list_ignore_vars
Output:	List of k clusters	

```
1. clusters [] = init_clusters (k, data) #randomly select k observations as centroids
2. iteration = 0
3. while iteration <= max_iter do
4.     for i = 0 to data.observations.length. - 1 do
5.         obs = data.get_observation (i)
6.         assign_cluster_near_by (clusters, obs, dis_type, list_ignore_vars)
7.     end for
8.     for i = 0 to k - 1 do
9.         recalculate_centroid (clusters [i])
10.    end for
11.    iteration = iteration +1
12. end while
13. return clusters []
```

1 $\Sigma 1120$	2 <table><tr><td>Machining</td><td>120</td></tr><tr><td>Manual</td><td>80</td></tr></table>	Machining	120	Manual	80	3 <table><tr><td>Yes</td><td>99</td></tr><tr><td>No</td><td>101</td></tr></table>	Yes	99	No	101	4 <table><tr><td>PIONEER 30F32</td><td>37</td></tr><tr><td>DK 234</td><td>55</td></tr><tr><td>FNC 114</td><td>48</td></tr><tr><td>Other</td><td>60</td></tr></table>	PIONEER 30F32	37	DK 234	55	FNC 114	48	Other	60	...	115 $\Sigma 1020120$
Machining	120																				
Manual	80																				
Yes	99																				
No	101																				
PIONEER 30F32	37																				
DK 234	55																				
FNC 114	48																				
Other	60																				
$V = 5.6$	$V = \textit{Machining}$	$V = \textit{No}$	$V = \textit{Other}$		$V = 5100.6$																

Figure 4. Representation of a cluster centroid CLR approach.

2.5.2. Multi-Start Simulated Annealing (MSSA)

Multi-start simulated annealing (MSSA) metaheuristic was used as a search strategy in the solution space generated in the CLR approach. The operation of this metaheuristic applied to the problem is presented in Algorithm 2, and the relevant steps are explained below.

Initial Assignment: MSSA starts from an initial solution built in the initial shuffling phase (generate_solution_models method); this is taken as the best solution called “best_sol” (lines 1 to 2). The generate_solution_models() method internally generates the linear regression models for each cluster. These models are created using Weka’s linear regression class [48]. The configuration eliminates attributes that exhibit collinearities, performs attribute selection to keep the variables most related to the target variable, and uses Ridge regularization to reduce model complexity and prevent overfitting. In addition, the statistical significance value (*p*-value) must be less than 0.05 for the construction of the models, which ensures that the relationship between the dependent variable and the independent variables is statistically significant. Thus, the CLR model obtained guarantees that the models of each cluster follow a linear trend.

Algorithm 2. Adapting the MSSA Algorithm to the CLR Approach.

	Initial Solution:	list_of_clusters
	Dataset:	data
Input:	Total Time:	total_time
	Number of starts:	num_starts
	List of variables to ignore:	list_ignore_vars
Output:	Best Solution with Regression Models for K Clusters	

```
1.  s = generate_solution_models (list_of_clusters, data, list_ignore_vars)
2.  best_sol = s
3.  time_boot = total_time/num_starts
4.  boot = 0; min_t = 0.05; max_t = 1
5.  e = exp ((log (min_t) – log (max_t))/time_boot)
6.  while boot < num_starts do
7.      t = 1
8.      initial_time = current_time_system ()
9.      time_release = 0
10.     while time_release < time_boot do
11.         list_of_clusters2 = get_neighbor (list_of_clusters)
12.         s1 = generate_solution_models (list_of_clusters2, data, list_ignore_vars)
13.         if is_a_valid_solution (s1) then
14.             if is_better (s, s1) o random (0, 1) < exp ((f(s) – f(s1))/t) then
15.                 list_of_clusters = list_of_clusters2
16.                 s = s1
17.                 if is_better (best_sol, s) then
18.                     best_sol = s
19.                 end if
20.             end if
21.         end if
22.         t = t × e
23.         time_release = current_time_system () – initial_time
24.     end while[after*=]
25. end while
26. return best_sol
```

Iterative Process: Multiple restarts (lines 6–25) are performed to find a new solution $s1$ called a neighboring solution generated from the current solution s ; if this solution $s1$ (line 11) has better fitness than s , it replaces it. Otherwise, a random number is generated with a uniform distribution between 0 and 1, and if it is lower than the value of the Boltzman distribution (Line 14), it replaces it (lines 15 and 16). It is then compared if the new solution s has better fitness than $best_sol$ (line 17); if that is the case, s becomes the new $best_sol$ (line 18). Finally, it is updated using a geometric factor (line 22), and when the stop criterion (execution time) is met, the best solution “ $best_sol$ ” is returned.

The initial solution is an evaluation of the clusters generated by the k-means algorithm; this solution has a structure, as shown in Figure 3, where a multiple linear regression model is built for each cluster and is associated with a fitness value according to the K models of the solution. The stop criterion of this metaheuristic was established at a specific time for the entire process, which was divided into the number of starts (num_starts parameter) to leave local optimums in the search process. Considering the above, 60 min are defined for

executing the metaheuristic with four starts; the time per start will be 15 min, during which the temperature t varies according to the function defined in Equation (2).

$$t = t \times e^{\frac{\log 0.05 - \log 1}{\text{TimeBoot}}} \quad (2)$$

Equation (2) generates a geometric decrease in temperature, and its behavior is shown in Figure 5. It should be noted that temperature plays an essential role in the metaheuristics of Simulated Annealing as a strategy to get out of local optima because when the temperature takes values close to the maximum temperature (max_t), there is a greater probability of accepting solutions that are not efficient, that is, performing exploration, but when this probability decreases and tends to the minimum value (min_t), exploitation is carried out and only solutions better than the current solution are accepted.

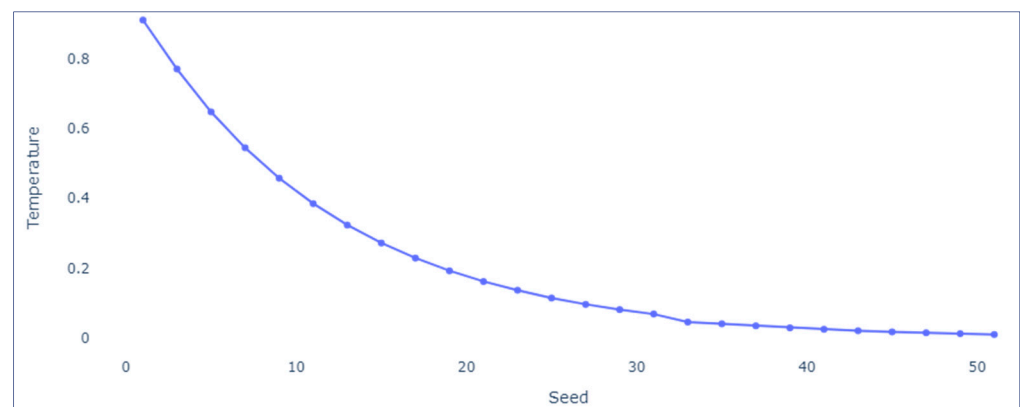


Figure 5. Temperature Behavior in MSSA.

According to Algorithm 2, the process of getting a neighbor (line 11) creates a new list of clusters (list_of_clusters2) generated from some changes made to a copy of the list of clusters (list_of_clusters) of the current solution. The changes made for the generation of a new cluster are described below:

Step 1: 5% of the observations are randomly chosen from each cluster and entered into a list of lists to be considered at the end of this step. Suppose that the 799 observations are distributed in 5 clusters, and each cluster is assigned the following numbers of observations [180, 159, 175, 160, 125]. Considering the above, a list with five temporary lists is defined with the following number of observations [9, 8, 9, 8, 7], which are randomly chosen from each cluster.

Step 2: Each temporary list in the list is traversed, and each observation is randomly assigned to a cluster different from the one from which it was extracted.

The new list of clusters is used to construct a new solution of the CLR approach; this new solution to be accepted must satisfy the feasibility criteria to make the new solutions s_1 generated from s valid solutions (feasible) even if they are not better. The comparison of the new solution s_1 with the current solution s , to determine if it is better, is conducted considering the models of each solution; this means that apart from the fitness of the solution, the lower R^2 adjusted of each solution is compared, and with this, it is ensured that the models of each solution have less variance. This comparison process is explained below with an example.

A new solution, s_1 , is remarkably like the current solution s if your fitness is around ± 0.1 or a parameter that can be set previously. If s_1 has fitness in this range, s y s_1 are considered similar, and a new comparison criterion is added, selecting as the best solution the one with the highest R^2 adjusted value of the worst cluster of each solution (the largest of the smallest).

Considering the range mentioned for the solution s , as seen in Figure 6, has a fitness of 0.799, the range would be from 0.679 to 0.879. As the new solution s_1 is within the

range established to be able to apply the second comparison criterion, the values of the models with lower R^2 adjusted for each solution are considered. The smallest R^2 adjusted of solution s corresponds to cluster $k1$ with 0.79, and that of solution $s1$ corresponds to cluster $k2$ with 0.772; therefore, when comparing R^2 adjusted, it is defined that $s1$ is a better solution than s and thus ensures that $s1$ has models with more homogeneous fits around the mean (fitness).

S		S1	
K1	K2	K1	K2
1 10 800 .. 51	4 720 8 .. 32	1 10 800 .. 51	4 720 8 .. 32
$y = a + b_1X_1 + b_2X_2 \dots + b_pX_p$	$y = a + b_1X_1 + b_2X_2 + \dots + b_mX_m$	$y = a + b_1X_1 + b_2X_2 + \dots + b_pX_p$	$y = a + b_1X_1 + b_2X_2 + \dots + b_mX_m$
$R^2 \text{ adjusted} = 0.79$	$R^2 \text{ adjusted} = 0.858$	$R^2 \text{ adjusted} = 0.788$	$R^2 \text{ adjusted} = 0.772$
$R^2 = 0.79$	$R^2 = 0.84$	$R^2 = 0.834$	$R^2 = 0.81$
$P \text{ value} = 0.000125$	$P \text{ value} = 0.000134$	$P \text{ value} = 0.00014$	$P \text{ value} = 0.00012$
fitness = 0.779		fitness = 0.779	

Figure 6. Example of solutions with $K = 2$.

When the new solution $s1$ has a fitness value that is outside the mentioned range, the evaluation is made only by the fitness value. This means that solution $s1$ will be better than s , according to the first criterion, if its fitness value is higher than the value of the current solution, for example, 0.9 compared to 0.779.

Finally, if a neighboring solution $s1$ replaces the current solution s , this solution is compared to the best solution best_sol found during the iterative process. If this new current solution is better, it will become the new best_sol . The best solution (best_sol) is the one that is returned when the stop criterion is reached.

2.5.3. GRASP

The GRASP metaheuristic was used similarly to MSSA to support the CLR approach to finding better solutions. Its operation is presented in Algorithm 3 and explained in detail below. GRASP is composed of two phases, which were adapted to solve the CLR problem as follows:

- **Phase 1—Construction:** The construction of a solution uses a greedy and intelligent function (line 6) in charge of creating an initial solution (list_of_clusters) generated by the k-means algorithm and selecting from each cluster the 20% of the observations that were furthest from the centroid, and this selection chooses 20% randomly to make an exchange to other clusters.

Figure 7 shows an example of a cluster with 30 observations, represented by green and blue circles. The yellow circle is the centroid of the cluster. Considering this illustration, the six blue observations correspond to the 20% of the observations furthest from the centroid, of which 20% are chosen randomly, in this case, one, and exchanged to another cluster.

When selecting observations farthest from the centroids, they are extracted from each cluster and entered in a temporary list, after the centroid of the affected cluster is recalculated. The observation exchange process is executed immediately after the observation selection from all the clusters. The choice of the target cluster of each observation is made by the mixed Euclidean distance to determine the most similar cluster, excluding the source cluster. Like removing an observation from one cluster, assigning it to another recalculates the affected cluster's centroid. At the end of the process, a new list of clusters was obtained and validated with the feasibility criteria. In this way, it is ensured that the construction phase obtains a viable solution to be optimized in phase 2.

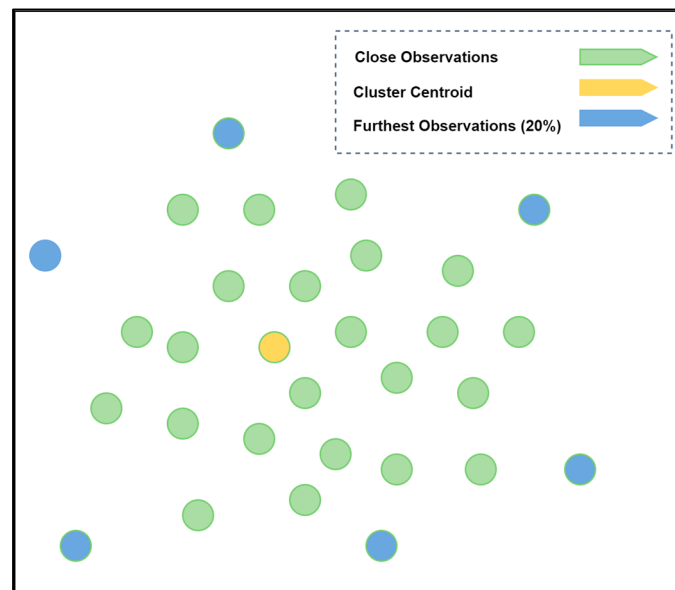


Figure 7. Example data distribution within a cluster GRASP metaheuristic.

Algorithm 3. GRASP algorithm adapted to the CLR approach.

Input:	Initial Solution:	list_of_clusters
	Dataset:	data
	Total Time:	total_time
	Maximum Iterations:	max_iter
	Fraction of time:	ft
	List of variables to ignore:	list_ignore_vars
Output:	Best solution with regression Models for K Clusters	

```

1.  best_sol = empty
2.  max_time = total_time/ft
3.  initial_time = current_time_system ()
4.  time_release = 0
5.  while time_release < total_time do
6.      listCluster2 = greedy_random_construction (list_of_clusters, data, list_ignore_vars)
7.      if best_sol == empty then
8.          best_sol = generate_solution_models (list_of_clusters2, data, list_ignore_vars)
9.      end if
10.     s = local_search (list_of_clusters2, data, list_ignore_vars, max_time, max_iter)
11.     if is_better (best_sol, s) then
12.         best_sol = s
13.     end if
14.     time_release = current_time_system () – initial_time
15. end while
16. return best_sol

```

- Phase 2—Optimization: Local optimization takes as its starting point the list of clusters built in phase 1 to find a better solution in a specific time that corresponds to a fraction of the total execution time of the GRASP metaheuristic. The operation of the local search is presented in Algorithm 4.

Algorithm 4. Local search applied in GRASP.

Input:	Initial Solution:	list_of_clusters
	Dataset:	data
	Total Time:	total_time
	Maximum Iterations:	max_iter
	List of variables to ignore:	list_ignore_vars
Output:	Best solution with regression models for K clusters	

```
1.  best_sol = generate_solution_models (list_of_clusters, data, list_ignore_vars)
2.  s = empty
3.  initial_time = current_time_system ()
4.  time_release = 0
5.  count = 0
6.  max_time = total_time/ft
7.  while time_release < total_time && stop_criterion != false do
8.      list_of_clusters2 = get_neighbor(list_of_clusters)
9.      s = generate_solution_models (list_of_clusters2, data, list_ignore_vars)
10.     if is_a_valid_solution (s) then
11.         if is_better (best_sol, s) then
12.             list_of_clusters = list_of_clusters2
13.             best_sol = s
14.             count = 0
15.         else
16.             count = count + 1
17.         end if
18.     else
19.         count = count + 1
20.     end if
21.     if count >= max_iter then
22.         break while
23.     end if
24.     time_release = current_time_system () – initial_time
25. end while
26. return best_sol
```

The local search initially takes best_sol as the solution generated from the list of clusters built in phase 1. It is followed by an iterative process of searching for a better solution within the neighborhood, with a maximum time (line 7) as a stop criterion or convergence where the algorithm does not find solutions with higher fitness than the best solution after a predefined number of iterations.

The process of obtaining a neighboring solution from a current solution is the same as that used in the MSSA metaheuristic (line 11), like the feasibility criteria for the validation and evaluation of constructed solutions. Finally, after executing the two phases described above, the best solution (best_sol) is returned when the maximum execution time is reached.

Figure 8 illustrates the CLR algorithm's workflow. Phase 1 involves building the mineable view using different variables (climate data, soil, and agricultural management practices) and several transformations, resulting in a final dataset with 116 predictor variables. In phase 2, the "Initial Shuffling" process takes the minable view and, using the k-means algorithm, defines k groups of observations with similar characteristics. For this study, the best results were obtained with k = 5. In phase 3, GRASP or MSSA metaheuristics

optimize the groups defined in the previous phase; this solution contains the groups of observations and their linear regression models that best predict each group's yield. Finally, in phase 4, the most relevant characteristics of the obtained CLR model are analyzed, and the quality of the model is compared to other state-of-the-art algorithms.

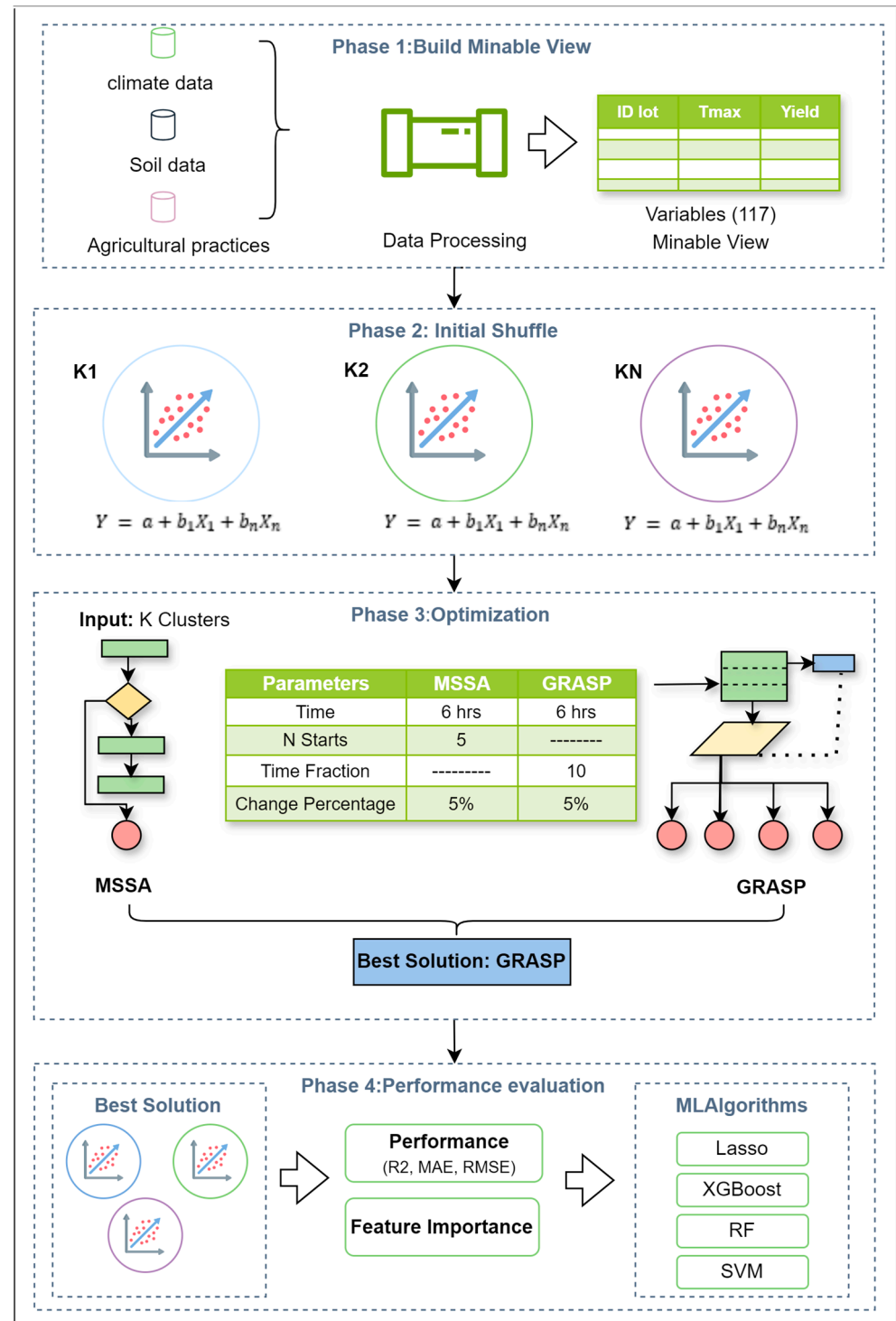


Figure 8. Flowchart of the implementation of the CLR algorithm for mazie yield prediction.

Finally, Table 7 presents a summary of modifications made to various artifacts, including metrics, algorithms, and metaheuristics for the construction of the CLR approach.

Table 7. Summary CLR approach artifact modifications.

Artifact	Description of Changes
Euclidean distance	This metric was adapted to make the calculation in mixed variables (continuous, nominal, and binary). For continuous variables, the difference between the variable associated with the observation and the variable associated with the centroid is calculated. For nominal and binary variables, we conducted special analysis to compare categories (numbers or strings). If the value of the variable is equal to the value of the centroid the distance is 0, if the value of the variable is different from the value of the centroid the distance is 1.
K-means algorithm	The centroid structure was modified to optimize the exchange of observations between clusters and reduce the number of operations. The centroid stores a control structure that consists of the following characteristics. In continuous variables, the total sum of the cluster observations and the average are calculated. In the nominal and binary variables, there is a count of the occurrences of the values by category of each observation.
MSSA Metaheuristic	The search criterion of metaheuristics is defined by a time T. The adaptation of MSSA consists of dividing the time into n starts, where the temperature function is reset at each start, avoiding falling into local optima. In the process of finding new solutions, clusters exchange observations and performance by cluster is calculated. The objective function computes a multiple linear regression for each cluster and obtains the R ² adjusted metric. Then, the average of the R ² adjusted values for all clusters is calculated (fitness). The best solution is the one with the highest average R ² adjusted value.
GRASP Metaheuristic	The process of adapting the metaheuristic involves generating a list of potential solutions. This list is built using the K-means algorithm, where a certain percentage of observations from each cluster are swapped with observations from other clusters. The observations selected for exchange are the ones farthest from the centroids, chosen at random. In local search, the objective function to maximize is the R ² adjusted average (fitness value) associated with the R ² adjusted of the different clusters. Another modification to the GRASP metaheuristic is the stop criterion: if a better candidate solution is not found after n iterations than the current one, the local search ends.

2.5.4. Optimization Function for MSSA and GRASP Metaheuristics

The adjusted coefficient of determination (Adjusted R²) is the quality metric used to evaluate each cluster's performance. Equation (3) defines this measure.

$$R_{Adjusted}^2 = 1 - \frac{N - 1}{N - nv - 1} \times (1 - R^2) \quad (3)$$

where N is the number of observations belonging to a cluster and nv is the number of variables selected in the model for the cluster. Considering that each linear regression model has an associated Adjusted R² as a quality metric, the CLR model as a model conformed by k clusters defines its quality as the average of the Adjusted R² associated with its corresponding linear models (see Equation (4)).

$$fitness = \frac{\sum_{i=1}^k R_{Adjusted}^2}{k} \quad (4)$$

where k is the number of clusters.

A solution is valid if all its clusters meet the following restrictions: (1) each associated linear regression model has a p -value less than 0.005, and (2) the number of observations is greater than the number of predictor variables.

3. Results and Discussion

3.1. Test Scenario

The K-Means algorithm was used to construct the initial groups, testing with k values from 2 to 5. Given the dataset's high dimensionality (116 variables), the algorithm was run for 500 iterations, and the mixed Euclidean distance was used as a metric to measure the similarity between instances.

The MSSA metaheuristic was initially run for 3 h for different values of k , from 2 to 5. Considering that the best results were achieved with $k = 5$, it was run again for 6 h to obtain better models. Table 8 summarizes the parameters used in the experimentation. The same steps were executed for the GRASP metaheuristic (see Table 9).

Table 8. Hyperparameter values for MSSA.

Variable	Value
Total Time (milliseconds)	time = 10,800,000 ... time = 21,600,000
Number of Starts	nBoot = 5c
List of variables to ignore	listIgno = [Rdt_Adjusted, Id_Lot]
Percentage of changes	Pchanges = 0.05

Table 9. Hyperparameter values for GRASP.

Variable	Value
Total Time (milliseconds)	time = 10,800,000 ... time = 21,600,000
Fraction time	Ft = 10
List of variables to ignore	listIgno = [Rdt_Adjusted, Id_Lot]
Percentage of changes	Pchanges = 0.05
Maximum iterations	mi = 20

3.2. Comparison of Results

Each algorithm (MSSA, GRASP) was executed 30 times (appropriate value to have a mean and deviation that tends to have a normal distribution according to the central limit theorem), with 2, 3, 4, and 5 clusters. Once the tests were executed, the average R^2 adjusted by the number of clusters was calculated; the results are shown in Table 10. The two adaptations of MSSA and GRASP found a higher average R^2 adjusted when K was equal to 5.

Table 10. Average of R^2 adjusted with 2, 3, 4, and 5 clusters with 30 independent repetitions.

K	MSSA	GRASP
2	0.768526008	0.779104195
3	0.800398698	0.814187886
4	0.832382658	0.844995661
5	0.859444174	0.862710015

3.3. Comparison of Solutions Using Different Values of K

To identify whether either of the two proposed solutions performs better, Wilcoxon's non-parametric statistical test is conducted for two paired samples to assess if a statistically significant difference exists between the means of the results obtained. The Wilcoxon test was designed to compare two samples when the data distribution cannot be defined a priori. The test seeks to know whether the differences between the two samples are due to chance.

According to the results of Wilcoxon's test on the values in Table 10, there was no significant difference between the means of the R^2 adjusted obtained in the regressions of

the proposed solutions. Therefore, it is impossible to determine whether the adaptation developed with GRASP or MSSA performs better.

3.4. Comparison of Solutions with Five Clusters

Not finding an adaptation with superior performance and considering that the best solution obtained was with 5 clusters ($k = 5$), each of the executions is analyzed to identify if there is a statistically significant difference within the executions. The Wilcoxon test has as its parametric alternative the t -test for paired samples. Since there are 30 values, a result of 30 new runs for $K = 5$ with twice the time, 6 h, these two tests were used to assess the difference between the means of the samples.

Figure 9 shows the result of the thirty (30) independent executions for MSSA and GRASP metaheuristics. When comparing the R^2 adjusted average values using the Wilcoxon and t -tests, it was found that there is no statistically significant difference for MSSA and GRASP solutions with $K = 5$ clusters.



Figure 9. Average R^2 adjusted for 5 clusters with 6 h of execution per result.

3.5. Selecting the Best CLR Model

Finding no statistically significant difference between the quality measures of both proposed solutions and with the need to define the best CLR model (solution with five centroids and its corresponding regression models, one for each cluster), the ten best average R^2 adjusted were selected from Figure 9 among these ten models; there is no difference of more than 0.01% in quality, and the simplest CLR model is chosen because a model with fewer variables is more straightforward to interpret. Table 11 details the top 10 models found and the model selected.

According to the results of Table 11, the CLR model with the lowest average of selected variables was created by the GRASP metaheuristic with seed 41. This model was selected as the best CLR model for maize yield prediction using the previously presented data.

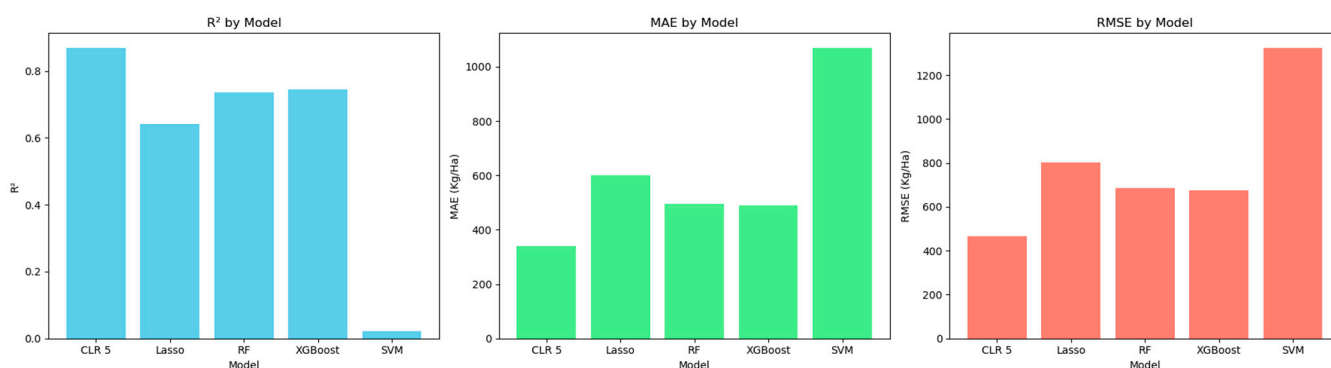
In summary, MSSA and GRASP were adapted and tested in the research process to determine the best metaheuristic for the adaptation of the CLR approach. Although one metaheuristic is not clearly superior to the other, the authors recommend GRASP for slightly better results and linear models with fewer variables.

Table 11. Top 10 Best average R^2 adjusted, most straightforward model selection.

Features			Number of Variables Selected per Cluster					Average Variables
Proposed Solution	Seed	Average R^2 Adjusted	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5	
GRASP	7	0.87890047	68	64	78	54	55	63.8
MSSA	59	0.87883411	60	79	61	35	72	61.4
GRASP	33	0.87750622	67	59	57	74	61	63.6
GRASP	31	0.87667793	49	42	62	65	65	56.6
MSSA	17	0.87643251	51	73	67	47	67	61
GRASP	41	0.87622102	46	68	48	59	57	55.6
GRASP	55	0.87572715	49	51	71	80	68	63.8
MSSA	33	0.87455223	55	55	66	59	73	61.6
MSSA	11	0.87399456	55	65	74	49	67	62
GRASP	51	0.87383793	54	70	66	72	47	61.8

3.6. Comparison with Other State-of-the-Art Algorithms

The best CLR model obtained with GRASP was compared against the models obtained by other machine learning algorithms, Lasso Regression, XGboost, Random Forest, and SVM, which used the same dataset. The dataset was divided into two sets: a training set (80%) and a test set (20%). The results for these models are shown in Figure 10, where the R^2 (higher values are better), MAE (lower values are better), and RMSE (lower values are better) metrics are evaluated.

**Figure 10.** Performance of linear and nonlinear models vs. 5-cluster CLR model.

Taking as a reference the performance of the Lasso linear model ($R^2 = 0.64$, RMSE = 802.48 kg/ha, and MAE = 600.35 kg/ha) and comparing it with the performance of the CLR 5 model ($R^2 = 0.87$, RMSE = 465.5 kg/ha, and MAE = 340 kg/ha), it is observed that CLR achieves better results in all metrics; likewise, if more complex models such as XGboost are compared, which achieves an acceptable performance ($R^2 = 0.75$, RMSE = 675.18 kg/ha, and MAE = 490.57 kg/ha), it is observed that CLR still maintains better performance (16% better in R^2 , 45% better in RSME, and 30% better in MAE). CLR obtains the best results in all metrics.

Although the results of research work using different data are not comparable, it is significant to mention that in [49], the authors implemented different ML algorithms (lasso, random forest, support vector, extreme gradient boosting, BP neural network, among others), to predict maize yield in China, making use of data with different variables (climate, satellite, and soil data). The obtained results show that the nonlinear models (support vector and extreme gradient boosting) are the ones that achieve the best performance ($R^2 \geq 0.75$, RMSE = 824–875 kg/ha, and MAE = 626–651 kg/ha). In comparison with the results obtained by the 5-cluster CLR model ($R^2 = 0.87$, RMSE = 465.5 kg/ha, and

MAE = 340 kg/ha), those results are lower. On the other hand, in [50], the researchers use Random Forest and artificial neural networks to predict crop yield, achieving an accuracy of 95%. However, these algorithms operate as a black box, making interpreting the results and decision-making difficult. Unlike the CLR model obtained, which is based on linear regression models that are naturally interpretable [51], it allows farmers or experts to understand the impact of the predictors on the response variable.

3.7. Characterization of Each Cluster in the Best Model

Table 12 shows the number of records for each group and the correlation coefficients of the linear regression models for each group in the best CLR model obtained. As can be seen, each group has between 137 and 204 instances, and the coefficients of determination (R^2) for each group are above 90%; this ensures that when classifying a new instance in the appropriate group, the prediction model will be precise in predicting performance.

Table 12. clusters characterization in the 5-cluster CLR model.

Metric	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5
Records Number	147	204	165	146	137
R^2	0.9031	0.9167	0.9098	0.9553	0.9134
Adjusted R^2	0.85848	0.87476	0.87241	0.92463	0.85084

Of the 116 variables, those that most represent each cluster based on their centroid data are presented below. To select the most relevant variables, first the normalized standard deviation of all variables in the five centroids was calculated, and then the five variables with the greatest variability, i.e., those that make a centroid more different from the others, were selected.

Figure 11 shows the climatic variables with the highest incidence in each cluster. In this case, cluster 1 shows higher temperatures in the maturation (Temp_Avg_Mat) and formation (Temp_Avg_For) stages, and cluster 4 shows higher precipitation in the vegetation and maturation stages (Rain_Accu_Veg, Rain_Accu_Mat), as opposed to cluster 1, where there is no precipitation in the vegetation and maturation stages. It is observed that there are significant differences in the variables of temperature, precipitation, and solar energy in each of the crop stages, which influence each cluster differently.

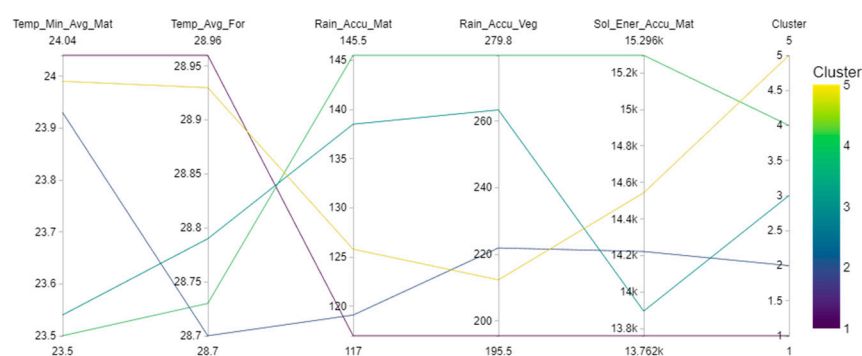


Figure 11. Climatic variables that further distinguish the clusters.

Figure 12 shows the characteristics related to the soil, where each cluster has different textures associated with it. In cluster 1, clayey (Porc_A) and clay loam (Porc_FAr) textures are predominant. Cluster 2 has a primarily soft texture (Porc_Soft). In cluster 3, the silty clayey texture (Porc_ArL) has a greater presence, while in cluster 4, the clay loam texture predominates (Porc_FAr). Finally, cluster 5 has non-soft soil with clayey (Porc_A) and clay loam (Porc_FAr) textures. In this context, identifying the type of texture of each cluster allows for understanding the differences in water retention capacity, nutrient availability,

and drainage behavior of the soil, thus allowing the management of appropriate strategies for agricultural management.

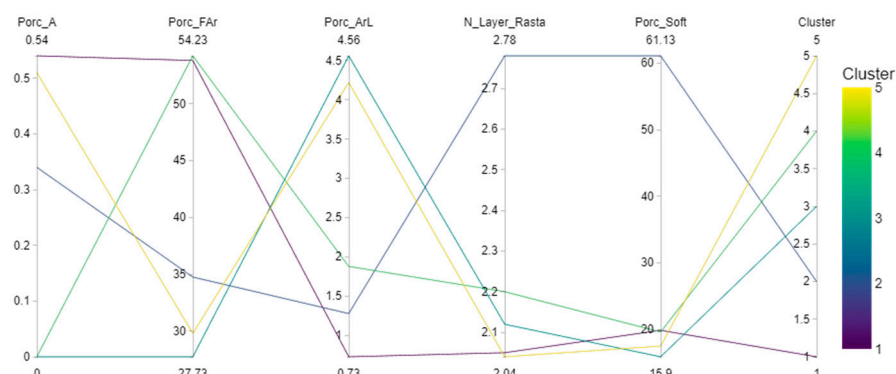


Figure 12. Soil variables that further distinguish the clusters.

Figure 13 shows the incidence of different management practices within each cluster. It can be observed that only in clusters 3 and 4 was mechanized weed control carried out in the flowering-harvest stages (MecWeedCont_Flow_Har). It is also observed that in clusters 1, 3, 4, and 5, pest control using chemicals was conducted in the emergence-flowering stages (PesContChem_Emer_Flow), but for cluster 2, no. In addition, only in cluster 1 phosphorus was applied before sowing (TotP_Before_Sow). On the other hand, organic fertilization was applied only in cluster 3 in the emergence-flowering stages (FerOrg_Emer_Flow) and not in the others. Finally, potassium was applied only in clusters 1 and 3 before sowing (TotK_Before_Flow).

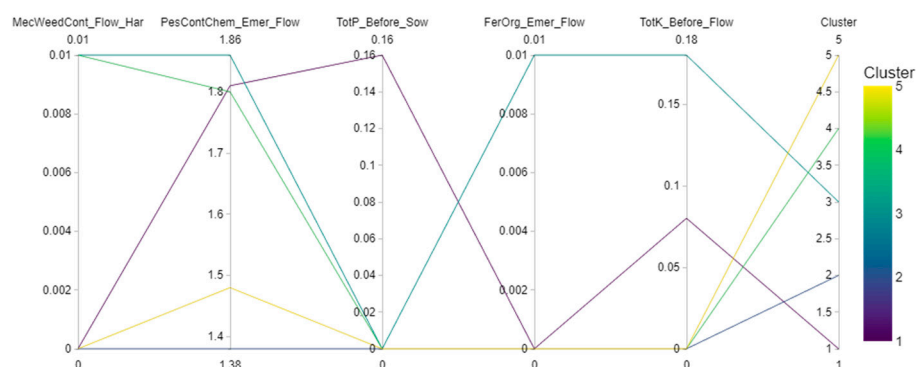


Figure 13. Management practices variables that further distinguish the clusters.

Next, Tables 13–17 describe in more detail the variables that differentiate each cluster. Each table shows first soil variables (characteristics), then climate variables, and finally, management practices variables (handling).

From the characterization of the clusters, it was possible to conclude that those with the best average yield are associated with drainage and good fertilization; in the case of the worst average yields, there is a relationship between heavy and poorly drained soils. The distribution of observations within the five clusters obtained confirms what was stated in site-specific agriculture (SSA), which indicates that two or more crop events are similar not in their geographical proximity but in their agricultural management practices, climate, and soil characteristics.

Table 13. Descriptive analysis of cluster model 1.

Cluster 1 (Third Best Yield Average “4768 kg/ha”)
Soil characteristics:
• Granular structure. • Spaced vegetal coating. • Good internal drainage. • Slow external drainage. • The soil mostly has a Far texture (53.82%) and Friable hardness (45.73%).
Climate characteristics:
• It has the highest frequency of days with maximum temperature above 34 °C in all its stages (40% in the vegetative and formation stages; 30% in the maturation stage); in addition to the interior of the cluster, the highest frequency of days with maximum temperature above 34 °C is found in the formation stage (78%). • Within the cluster, the highest accumulated energy in the vegetative stage (16,960) and the lowest accumulated energy in the formation stage (19,463) were reported. • It has higher diurnal range values in degrees Celsius in the vegetative and formation stages (9.1, 9.5, and 9.14).
Handling characteristics:
• Only clusters where N, P, and K were applied before planting. • Drainage is conducted for the crop. • About 3/4 of the records performed mechanized seeding. • About 2/3 of the records showed cotton as a previous crop.

Table 14. Descriptive analysis of cluster model 2.

Cluster 2 (Fourth Best Average Yield “4568 kg/ha”)
Soil characteristics:
• Granular structure. • Regular plant coating. • Slow to very slow internal drainage. • No external drainage. • It has a lower adequate depth (32 cm, 12 cm lower than the average of the other clusters). • Speckling is observed, and the average depth is 18.82 cm. • The soil mostly has a Far (34.75%), Ar (25.57%), and Soft (61.13%) texture.
Climate characteristics:
• It has the lowest frequency of days with maximum temperature above 34 °C in all its stages (30% in the vegetative and formation stages; 22% in the maturation stage). In addition, like cluster 1, it has the highest frequency of days with a maximum temperature above 34 °C in the formation stage (78%). It also stands out because it is found at the highest frequency of days in the maturation stage (91%). • The average accumulated precipitation in mm is among the lowest during the three stages (221, 247, 119). Within the cluster, the lowest accumulated precipitation in mm was reported for the vegetative stage (51.4) and the formation stage (14.9). • The average precipitation frequency below 10 mm is among the lowest for the three stages (16%, 14%, and 10%). • It has the highest average humidity during the three stages (83%, 82%, 83%). Within the cluster, the lowest accumulated energy in the vegetative stage (13,132) and the highest accumulated energy in the maturation stage (27,398) were reported. • It has lower diurnal range values in degrees Celsius in the three stages (8.71, 9.12, and 8.77).
Handling characteristics:
• The least amount of fertilizer was applied overall. • Drainage is not suitable for cultivation. • About 2/3 of the records perform mechanized seeding. • About 1/2 of the records showed cotton as a previous crop.

Table 15. Descriptive analysis of cluster model 3.

Cluster 3 (Best Average Yield “5102 kg/ha”)
Soil characteristics:
• Lumpy structure. • Good plant coating. • Good internal drainage. • No external drainage. • The soil mostly has an Ar texture (41.85%) and firm hardness (40.94%).
Climate characteristics:
• The vegetative and maturation stages have among the highest frequency of days with maximum temperature 34 °C (36% and 34%), but the formation stage is among the lowest (20%). • The average accumulated precipitation in mm is among the highest for the three stages (263, 256, 138). • The frequency of days with more than 10 mm of precipitation is among the highest for the three stages (18%, 15%, and 13%).
Handling characteristics:
• It has the highest amounts of N and K applied before planting. • The only cluster where organic fertilization is reported (between the emergence and flowering stages). • Adequate drainage for cultivation. • About 3/4 of the records perform mechanized seeding. • About 2/3 of the records showed cotton as a previous crop.

Table 16. Descriptive analysis of cluster model 4.

Cluster 4 (Second Best Average Yield “4902 kg/ha”)
Soil characteristics:
• A lumpy structure. • Good plant coating. • Leaf litter is observed. • Good internal drainage. • Slow external drainage. • The soil mostly has a Far texture (54.23%), Firm hardness (34.11%), and Friable (33.4%).
Climate characteristics:
• The cluster accumulated the most energy in the formation stage (26199) and the least in the maturation stage (2996). • It has a higher daytime range value in degrees Celsius (9.43) in the maturation stage.
Handling characteristics:
• Adequate drainage for cultivation. • About 3/4 of the records perform mechanized seeding. • About 1/2 of the records showed maize as a previous crop.

Table 17. Descriptive analysis of cluster model 5.

Cluster 5 (Worst Average Yield “4335 kg/ha”)
Soil characteristics:
• Lumpy structure. • Good plant coating. • Good internal drainage. • No external drainage. • The soil mostly has a texture FAr (29.79%), Ar (23.59%), and hardness Firm (31.75%), Lasted (26.86%).

Table 17. Cont.

Cluster 5 (Worst Average Yield “4335 kg/ha”)
Climate characteristics:
• Within the cluster, the highest frequency of days with maximum temperature above 34 °C was reported in the vegetative stage (82%). • The average accumulated precipitation in mm is among the lowest for the three stages (212, 245, 125). • It has the lowest average humidity during all three stages (80%, 80%, and 81%). • Within the cluster, the lowest accumulated energy was reported in the vegetative stage (13,132).
Handling characteristics:
• Adequate drainage for cultivation. • About 1/2 of the records do manual seeding. • About 1/2 of the records showed maize as a previous crop.

When analyzing the models obtained, various characteristics that impact each cluster are identified. For instance, the number of chemical weed controls conducted between the planting and emergence stages of the plants was influential in only one of the five clusters. On the other hand, the lot height (lot_height) was found to be important in four out of the five clusters (Cluster 2, Cluster 3, Cluster 4, Cluster 5), with each centroid having a different average height value. These values were 9.48, 12.48, 11.66, and 12.69, and the correlation of this variable with the yield was 0.14, 0.56, 0.11, and -0.12 , respectively. The incidence of this variable in four clusters is an essential factor in SSA because it helps to identify that the characteristics impact differently on each cluster’s production (performance). Figure 14 shows the correlation of the variable lot height (Lot_Height) with the variable crop yield (Yield) for clusters two, three, four, and five.



Figure 14. Correlation between lot height vs. yield maize. Cluster 2, Cluster 3, Cluster 4 and Cluster 5.

Additionally, in the five clusters, the presence of the following variables was identified.

- The seed used in planting (Genetic_Material).
- Plant density at 20 days (Population_20Days_Adj).
- Intensity of plants with low production (Observe_Small_Plants_Rasta).
- Vegetation cover (Plant_Covering_Soil_Rasta).
- the presence of hard layers that can impede root growth, water movement, and soil respiration (Cap_Endure_Rasta).

The above indicates that these variables are relevant in any of the identified crop scenarios; however, the level of correlation between them and their impact on crop yield varies in each cluster, which again leads to the conclusion that the SSA needs to identify the different levels of importance and incidence within each cluster.

Considering the different multiple linear regression models obtained and the coefficients associated with the variables, the coefficients of the genetic material variable denote that the impact within each cluster is different; it also helped to identify that not all materials provide the same effect if used in all clusters, which is one of the critical factors in SSA. Table 18 shows the best options for genetic material by cluster, where farmers can choose whether their crop has soil and climate characteristics like the patterns identified in the five clusters.

Table 18. Genetic material by cluster in the best CLR model found.

Genetic Material	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5
PIONEER 30F35 H	X	X		X	
PIONEER 30F35 HRR	X	X			
ADV 9293 (Syngenta)		X	X		
P3966 (Pioneer)		X		X	
ADV 9339 (Syngenta)				X	X
ICA V 305	X				
PIONEER 30F32	X				
Sinko (Syngenta)	X				
DK7088		X			
PIONEER 30F32HW			X		
Impact (Syngenta)			X		
DK 1596			X		
Cerato (Syngenta)			X		
P4082 (Pioneer)				X	
Other ¹				X	
PIONEER 30F35					X
DK7088					X
DK 234					X
PAC 105					X

¹ Genetic material is not included in the list of options available in the FENALCE form within the SYRIA platform.

3.8. Analyzing the Coefficients of Linear Regression Models in Each Cluster

Table 19 shows the five most influential variables in the prediction of maize yield for each cluster; in this case, the coefficients of the respective linear regression models are analyzed. For Cluster 1, the most influential coefficients are related to crop practices such as Days to emerge (Days_to_Emerge), harvesting method (manual, mechanized), and weed controls using chemicals carried out in emergence and flowering stages (ContWeedsChem_Emer_Flow). For Cluster 2, the most influential features in the prediction are related to the climate, such as Humidity in the Vegetation and Formation stages (Rhum_Avg_Veg, Rhum_Avg_For), maximum temperatures in the formation stage (Temp_Max_Avg_For), and the chemical fertilizers applied between emergence and flowering stages (FerQui_Emer_Flow).

Table 19. Most influential coefficients of each regression model associated with each cluster.

Cluster Name	Features	Coefficients	Absolute Importance
Cluster 1	Days_to_Emerge	112.28	112.28
	Rhum_Avg_Form	−95.59	95.59
	Mechanized_Harvesting_Method	86.48	86.48
	Manual_Harvesting_Method	−86.48	86.48
	ContWeedsChem_Emer_Flow	83.29	83.29
Cluster 2	Rhum_Avg_For	−116.89	116.89
	Diurnal_Range_Avg_Veg	−108.06	108.06
	Temp_Max_Avg_For	86.48	86.48
	FerQui_Emer_Flow	82.38	82.38
	Rhum_Avg_Veg	80.39	80.39
Cluster 3	Lot Height	110.11	110.11
	ContPestChem_Emer_Flow	−95.40	95.40
	Genetic_Material_Dk 234	−79.44	79.44
	soil_structure_lumpy_rasta	77.15	77.15
	Manual_Harvest_Method	−70.32	70.32
Cluster 4	Previous_Cultivation_Grass	−90.02	90.02
	Rasta_Earring	83.69	83.69
	Thickness_Layer_Endure_Rasta	72.29	72.29
	Observe_White_Scasts_Rasta	62.28	62.28
	ContPestsChem_Flow_Harvest	61.24	61.24
Cluster 5	Rhum_Avg_Veg	115.25	115.25
	Rhum_Avg_For	−105.67	105.67
	FerQui_Emer_Flor	102.31	102.31
	Drainage_Yes	−82.45	82.45
	Drainage_No	82.45	82.45

On the other hand, in Cluster 3, the most influential variables are related to the lot height (Lot Height), pest control using chemicals between emergence and flowering stages (ContPestChem_Emer_Flow), seed genetic material (Genetic_Material_Dk 234), and the soil structure (soil_structure_lumpy_rasta). In cluster 4, the most influential coefficients in the prediction are related to soil features such as previous crop grass (Previous_Cultivation_Grass), the thickness of the Rasta soil layer (Thickness_Layer_Endure_Rasta), the absence of white crusts on the rasta (Observe_White_Scasts_Rasta), and pest control by chemicals in flowering to harvest stages (ContPestsChem_Flow_Harvest). Finally, in cluster 5, the most influential coefficients are related to climate variables such as humidity in vegetation and formation stages (Rhum_Avg_Veg, Rhum_Avg_For), chemical fertilization applied in emergence to flowering stages (FerQui_Emer_Flor), and soil drainage (Drainage_Yes, Drainage_No).

Based on the analysis presented above, it is evident that the variables that influence corn yield prediction differ from one cluster to another; therefore, specific strategies must be defined according to the local conditions of each farm based on the experience of similar farms.

Finally, it can be verified that the models obtained using the CLR approach achieve an R^2 of 87%. This is in comparison to other Machine Learning models such as Lasso linear regression, Random Forest, XGBoost, and Support Vector Machine (SCV), which achieve a maximum R^2 of 75%, where all observations are used as if they belonged to a single cluster (dataset), which demonstrates that the CLR approach proposed here allows obtaining a more accurate machine learning solution to predict the yield of a maize crop. In addition,

with this approach, it is possible to detect relationships that only exist in subsets of the total population (Clusters), which are not detectable by other Machine learning algorithms reported in the literature.

3.9. Limitations

To implement the proposed CLR algorithm, the following must be fulfilled:

- Like other machine learning algorithms, the data used must represent the problem to be solved, and there must be considerably more instances (observations) than variables (features), especially if there are many variables.
- The data cleaning and transformation process is not just a step but a crucial and meticulous task that must be conducted to obtain a quality and unbiased minable view.
- Since linear regression models are sensitive to outliers, they should be treated well in the construction phase of the minable view so as not to affect the construction of the CLR model. In this context, according to [52], recent work has been performed on CLR models strengthened in the presence of mild outliers in small subpopulations; this subject is a fascinating research topic for future versions of the present work.
- Attribute selection is not recommended for constructing the mineable view, since variables that may be irrelevant when viewing the data as a whole may be very relevant when divided into groups.
- The definition of the “ideal” number of groups (k) is not simple. It’s a complex process that demands significant computational time, as it involves evaluating the algorithm from a value of 2 to a maximum value that may be defined by time or computational resource constraints.

4. Conclusions and Future Work

4.1. Conclusions

This study improved the prediction of corn crop yield in the department of Córdoba, Colombia, using the Cluster Linear Regression (CLR) approach. After an optimization process using the MSSA or GRASP metaheuristic, five clusters were identified, each with its own linear regression model capable of predicting the target variable (yield). The proposed model employs characteristics related to climate, soil, and agricultural practices, reaching an R^2 of 87%, significantly outperforming other algorithms reported in the literature, such as Lasso linear regression, SVM, XGBoost, and Random Forest. The GRASP metaheuristic performs slightly better than MSSA and generates a simpler model for interpreting the results.

A detailed analysis of the most influential variables in each cluster revealed that the highest yields, with an average of “5102 kg/ha”, were associated with cluster 3. This cluster is characterized by mostly clay soils (Ar) with good internal drainage and slow external drainage. In addition, the highest application of nitrogen (N), phosphorus (P), potassium (K), and chemical fertilizers was made during the stages from emergence to flowering. The use of machinery for planting and harvesting was also observed. On the contrary, cluster 5 presented the lowest yield, with a mean of “4335 kg/ha”, and is characterized by the absence of external drainage and clay loam soils (FAr) of firm hardness. In this cluster, planting and harvesting practices are conducted manually.

From the analysis of the variables that influence each cluster, it is concluded that the most relevant factors for maize crop yield are the soil characteristics, such as resistance to breakage, texture, and external drainage. In addition, it was identified that using machined tools significantly improves performance. The differences identified through the CLR approach allow specific agricultural management recommendations and fertilization strategies adapted to each cluster to be designed, contributing to increasing productivity levels. Finally, the CLR algorithm proposed in this research performs better than other linear and nonlinear algorithms reported in the literature. However, its most significant advantage

is the interpretability of the results, which allows for the analysis and understanding of the contribution of each predictor in the response variable.

4.2. Future Work

About the definition of the mineable view, it was found that the clusters differed significantly in terms of soil texture and resistance to breakage. However, while dividing the variable containing this information into several columns with percentage values, each layer's original order of occurrence was not maintained (from the shallowest to the deepest). This is an essential factor for this characteristic because it would indicate which of these characteristics it would have contact with the plant thanks to the depth of its roots, so it is helpful that in future works, a way to preserve the order of these is proposed.

Future research should include the variable representing soil color (wet and dry) with an RGB representation. In this study, it was excluded because it had 54 different values, and considering would further increase the dimensionality of the mineable view. In addition, other variables, such as organic matter and drainage, could reduce the loss of information provided by this variable.

Regarding the CLR process, it would be recommended that additional research involving new metaheuristics be conducted to search for the best distribution of K clusters of crop events. It would also be helpful to define new strategies for identifying the neighborhood for a grouping of observations, which should include more significant knowledge of the problem as much as possible.

The large number of variables that affect a crop can have complex relationships that the CLR approach cannot detect; therefore, as future work, the analysis and exploration of the Clusterwise nonlinear regression (CNLR) approach is proposed, where nonlinear relationships are modeled within each cluster.

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